

Object Oriented Programming

Dr. Imran Ashraf

Office: 02

`imran.ashraf[at]hitecuni[dot]edu[dot]pk`

Review

- Pointers
- References
- Functions
 - Passing arguments to functions
 - Function Overloading

Today's Lecture

- Scope of variables
- Lifetime of variables
- Functions
 - Function Overloading

Scope of a variable

- Scope is the region or section of code where a variable can be accessed.
- Local scope
 - Function
 - Block
- Global scope

Quiz

- Variables should be declared in local or global scope?
- What will happen if you declare a variable with the same name in local as well as in global scope?

Life of a variable

- The lifetime of a variable or object is the time period in which the variable/object has valid memory.
- Examples:
 - Static: entire duration of the program's execution, allocated in data segment.
 - Automatic: block scope, allocated on stack.
 - Dynamic: from allocation to free, allocated on heap

Review: Function Signature

- **Function name**
- **Number of arguments**
- **Arguments' type**
- **Arguments' order**
- Arguments' name
- Return type

Function Overloading

- define two or more functions with the same name and in the same scope.
- Example:

```
int test() { }  
int test(int a) { }  
float test(double a) { }  
int test(int a, double b) { }
```


Why Function Overloading?

- Programming convenience
- A form of polymorphism (many forms?)

Why Function overloading? Contd.

- `function Person[] FindPersons(string nameOfPerson) { ... }`
 - `function Person[] FindPersons(date dateOfBirth) { ... }`
 - `function Person[] FindPersons(int age, string dogsName) { ... }`
-
- `function Person[] FindPersonsByName(string nameOfPerson) { ... }`
 - `function Person[] FindPersonsByDOB(date dateOfBirth) { ... }`
 - `function Person[] FindPersonsByAgeAndDogsName(int age, string dogsName) { ... }`

Quiz

- Write a simple program where function overloading can be used?

Quiz

- Write a program containing two functions with the same name as **absolute()**. The first function finds the absolute value of an integer and the second function finds the absolute value of a float. Write the **main()** to test these two functions.

Summary

- Scope of variable
- Life of variables
- Function overloading

Next Lecture

- Classes and Objects